

1882-1935

The mathematician Emmy Noether had a brilliantly creative mind. Her ability to make conceptual connections in a whole new way transformed the fields of abstract algebra and theoretical physics.

Amalie Emmy Noether was born in the German town of Erlangen. She gained a doctorate in mathematics but struggled with discrimination against women in academia. Working for years without pay, she had to teach under the name of a male colleague. Despite this, she succeeded in winning admiration for her work. She is best known for Noether's theorem, which connects two key concepts within physics: conservation laws and symmetries. Each law can be associated with a particular symmetry and vice versa. Noether was described by Albert Einstein as a "creative mathematical genius."

MILESTONES

FEMALE GRADUATE

Gains a doctorate in mathematics in 1907: only the second woman to receive that degree.

LANDMARK THEOREM

Proves her theorem in 1918, fundamental to both particle physics and general relativity.

GAINS RECOGNITION

Awarded the highly respected Ackermann-Teubner Memorial Prize in mathematics in 1932.

FLEES GERMANY

Forced by the Nazis to leave Germany in 1933. Moves to the United States and takes up a post there.

"My methods are really methods of **working and thinking."**

Emmy Noether, 1931

EMMY
NOETHER